

# How communication apprehension and publication transportation ride share differ by country

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## Topic

a visualization for “How communication apprehension and publication transportation ride share differ by country”

## Method

- 1. we use violin plot to visualize
- 2. add 12 columns for CA
- 3. add 4 columns for PTRS(public transportation ride share)

## workflow for generating the photo

import three .csv files

```
import pandas as pd

df_a = pd.read_csv("PRCAProlificExport_FileA.csv")
df_b = pd.read_csv("PRCAProlificExport_FileB.csv")
df_c = pd.read_csv("PRCAQualtricsExport_FileC.csv")

# stick df_a and df_b
df_1 = pd.concat([df_a, df_b], ignore_index=True)
print(df_1)
```

|     | Submission id \                                   |
|-----|---------------------------------------------------|
| 0   | c4ab54135f26292de9faf6a1043ba0817e8c2fb29cd050... |
| 1   | 91d238c88316a444585278049cc9bdedb871414ebdb0dd... |
| 2   | f3943bf594d541b81fbde88288600739a47ee97b0524c1... |
| 3   | 0799215a062ddef149fa49bfe069e3ff2b4b79f08f5b01... |
| 4   | f49783f27606316e1e3a590e7403eab417b5612c2faeaf... |
| ..  | ...                                               |
| 257 | a2d0b751f64d1d25c66dfc63b9b15e47d1ee91318f1bfd... |
| 258 | e1b1b90668b67a6539e9891be83e91a0dd305f010a7f5a... |
| 259 | 42dbbedc5431e2786ae049de6de4cf4225e289886e6a3e... |
| 260 | 5f1e0315552a4aa3890128761793b574646068215c236c... |
| 261 | a9095755df1f3d955b3619a9abfc2b354183e1c2c09cd3... |

|     | Participant id                                     | Started at \ |
|-----|----------------------------------------------------|--------------|
| 0   | ebab12e9878344fdc93a8c1cad27ff2f709e12b90f0496...  | 2025         |
| 1   | b6954799ca47bc836ac02ab86e4710c3b0e096d7ace449...  | 2025         |
| 2   | 6ac07c3845b0ce5cac4ab3569ea70236b7cfa8fd51576e...  | 2025         |
| 3   | c67b3266ecd4af12979c1f67d1efffd0d3e9bfbcb567d21... | 2025         |
| 4   | 1cefb8d34b92489e8ffd9cf34c469964f0527b1cae33e0...  | 2025         |
| ..  | ...                                                | ...          |
| 257 | bfbf5ad6de8d7d872d3ecbaacc2d0dac36436b0f3436a2...  | 2026         |
| 258 | f0fd850819db2cfaf10c1175fe69952b5061e43e99ed75...  | 2026         |
| 259 | cc86586976c616056a7a42cdc7acfdb68eb8dcc98ff968...  | 2026         |
| 260 | c7970d1139c4b82a0cb8829f720c302e300e95dabd004a...  | 2026         |
| 261 | 656258a8fcf5ec12a05a26fbcacb465a0cb4cfd467217f...  | 2026         |

|   | Completed at | Time taken | Age  | Total approvals | Sex \ |
|---|--------------|------------|------|-----------------|-------|
| 0 | 2025         | 188.0      | 36.0 | 700+            | Male  |

|     |      |       |        |         |                 |
|-----|------|-------|--------|---------|-----------------|
| 1   | 2025 | 110.0 | 41.0   | 700+    | Male            |
| 2   | 2025 | 63.0  | 39.0   | 700+    | Male            |
| 3   | 2025 | 85.0  | 61.0   | 700+    | Male            |
| 4   | 2025 | 142.0 | 45.0   | 700+    | Male            |
| ..  | ...  | ...   | ...    | ...     | ...             |
| 257 | 9999 | NaN   | 997.0  | 700+    | CONSENT_REVOKED |
| 258 | 9999 | NaN   | 1001.0 | 700+    | CONSENT_REVOKED |
| 259 | 9999 | NaN   | 999.0  | 500-599 | CONSENT_REVOKED |
| 260 | 9999 | NaN   | 1000.0 | 700+    | CONSENT_REVOKED |
| 261 | 9999 | NaN   | 999.0  | 700+    | CONSENT_REVOKED |

|     | Country of residence | Student status  | Employment status |
|-----|----------------------|-----------------|-------------------|
| 0   | United States        | Yes             | Part-Time         |
| 1   | United States        | No              | Other             |
| 2   | United States        | No              | Full-Time         |
| 3   | United Kingdom       | No              | Other             |
| 4   | United States        | No              | Full-Time         |
| ..  | ...                  | ...             | ...               |
| 257 | CONSENT_REVOKED      | CONSENT_REVOKED | Other             |
| 258 | CONSENT_REVOKED      | CONSENT_REVOKED | Other             |
| 259 | CONSENT_REVOKED      | CONSENT_REVOKED | Other             |
| 260 | CONSENT_REVOKED      | CONSENT_REVOKED | Other             |
| 261 | CONSENT_REVOKED      | CONSENT_REVOKED | Other             |

[262 rows x 11 columns]

```
# Merge the result with df_c using Q0 and participant id
df = pd.merge(
    df_1,
    df_c,
    left_on="Participant id",
    right_on="Q0",
    how="inner"
)

print(df.head())
```

|   | Submission id \                                   |
|---|---------------------------------------------------|
| 0 | c4ab54135f26292de9faf6a1043ba0817e8c2fb29cd050... |
| 1 | 91d238c88316a444585278049cc9bdedb871414ebdb0dd... |
| 2 | f3943bf594d541b81fbde88288600739a47ee97b0524c1... |
| 3 | 0799215a062ddef149fa49bfe069e3ff2b4b79f08f5b01... |

4 f49783f27606316e1e3a590e7403eab417b5612c2faeaf...

|   | Participant id                                    | Started at \ |
|---|---------------------------------------------------|--------------|
| 0 | ebab12e9878344fdc93a8c1cad27ff2f709e12b90f0496... | 2025         |
| 1 | b6954799ca47bc836ac02ab86e4710c3b0e096d7ace449... | 2025         |
| 2 | 6ac07c3845b0ce5cac4ab3569ea70236b7cfa8fd51576e... | 2025         |
| 3 | c67b3266ecd4af12979c1f67d1efffd0d3e9bfb567d21...  | 2025         |
| 4 | 1cefb8d34b92489e8ffd9cf34c469964f0527b1cae33e0... | 2025         |

|   | Completed at | Time taken | Age  | Total approvals | Sex  | Country of residence \ |
|---|--------------|------------|------|-----------------|------|------------------------|
| 0 | 2025         | 188.0      | 36.0 | 700+            | Male | United States          |
| 1 | 2025         | 110.0      | 41.0 | 700+            | Male | United States          |
| 2 | 2025         | 63.0       | 39.0 | 700+            | Male | United States          |
| 3 | 2025         | 85.0       | 61.0 | 700+            | Male | United Kingdom         |
| 4 | 2025         | 142.0      | 45.0 | 700+            | Male | United States          |

|   | Student status | ... | Q17                        | Q18 \             |
|---|----------------|-----|----------------------------|-------------------|
| 0 | Yes            | ... | Somewhat agree             | Somewhat agree    |
| 1 | No             | ... | Neither agree nor disagree | Somewhat disagree |
| 2 | No             | ... | Somewhat agree             | Somewhat disagree |
| 3 | No             | ... | Strongly agree             | Strongly disagree |
| 4 | No             | ... | Somewhat agree             | Strongly disagree |

|   | Q26              | Q27                        | Q28 \            |
|---|------------------|----------------------------|------------------|
| 0 | 2-4 days a month | 3-4 rides in a typical day | 0-1 days a month |
| 1 | Never            | 1-2 rides in a typical day | 0-1 days a month |
| 2 | Never            | 1-2 rides in a typical day | Never            |
| 3 | Never            | 1-2 rides in a typical day | Never            |
| 4 | Never            | 1-2 rides in a typical day | 2-4 days a month |

|   | Q29                        | Q20 | Q21 | Q18.1 | Q19 |
|---|----------------------------|-----|-----|-------|-----|
| 0 | 3-4 rides in a typical day | NaN | NaN | NaN   | NaN |
| 1 | 1-2 rides in a typical day | NaN | NaN | NaN   | NaN |
| 2 | 1-2 rides in a typical day | NaN | NaN | NaN   | NaN |
| 3 | 1-2 rides in a typical day | NaN | NaN | NaN   | NaN |
| 4 | 1-2 rides in a typical day | NaN | NaN | NaN   | NaN |

[5 rows x 38 columns]

save the merged file

```
df.to_csv("merged_file.csv", index=False)
```

read into the merged file

```
df = pd.read_csv("merged_file.csv")
```

```
## then code into numbers:
```

```
agree_scale = {  
    "Strongly disagree": 1,  
    "Somewhat disagree": 2,  
    "Neither agree nor disagree": 3,  
    "Somewhat agree": 4,  
    "Strongly agree": 5  
}
```

```
reverse_agree_scale = {  
    "Strongly disagree": 5,  
    "Somewhat disagree": 4,  
    "Neither agree nor disagree": 3,  
    "Somewhat agree": 2,  
    "Strongly agree": 1  
}
```

```
# Coding for Q26 and Q28
```

```
frequency_scale = {  
    "Never": 1,  
    "0-1 days a month": 2,  
    "2-4 days a month": 3,  
    "4-8 days a month": 4,  
    "8 or more days a month": 5  
}
```

```
#
```

```
scale_d = {  
    "1-2 rides in a typical day": 1,  
    "3-4 rides in a typical day": 2,  
    "5-6 rides in a typical day": 3,  
    "7 or more rides in a typical day": 4  
}
```

```
# Columns using the normal agreement scale
normal_cols = ["Q1", "Q3", "Q5", "Q13", "Q15", "Q18"]

# Columns using the reverse agreement scale
reverse_cols = ["Q2", "Q4", "Q6", "Q14", "Q16", "Q17"]

cols_c = ["Q26", "Q28"]

cols_d = ["Q27", "Q29"]

# Apply coding
df[normal_cols] = df[normal_cols].replace(agree_scale)
df[reverse_cols] = df[reverse_cols].replace(reverse_agree_scale)
df[cols_c] = df[cols_c].replace(frequency_scale)
df[cols_d] = df[cols_d].replace(scale_d)
```

```
df.head()
```

|   | Submission id                                     | Participant id                                    |
|---|---------------------------------------------------|---------------------------------------------------|
| 0 | c4ab54135f26292de9faf6a1043ba0817e8c2fb29cd050... | ebab12e9878344fdc93a8c1cad27ff2f709e12b90f0490... |
| 1 | 91d238c88316a444585278049cc9bdedb871414ebdb0dd... | b6954799ca47bc836ac02ab86e4710c3b0e096d7ace4...   |
| 2 | f3943bf594d541b81fbde88288600739a47ee97b0524c1... | 6ac07c3845b0ce5cac4ab3569ea70236b7cfa8fd51570...  |
| 3 | 0799215a062ddef149fa49bfe069e3ff2b4b79f08f5b01... | c67b3266ecd4af12979c1f67d1efffd0d3e9bfbc567d2...  |
| 4 | f49783f27606316e1e3a590e7403eab417b5612c2faeaf... | 1cefb8d34b92489e8ffd9cf34c469964f0527b1cae33e...  |

```
df["sum"] = (
    df[normal_cols].sum(axis=1) +
    df[reverse_cols].sum(axis=1)
)

df["PTRS"] = (
    df[["Q26", "Q28"]].sum(axis=1) +
    df[["Q27", "Q29"]].sum(axis=1)
)
```

```
df.head()
```

|   | Submission id                                     | Participant id                                    |
|---|---------------------------------------------------|---------------------------------------------------|
| 0 | c4ab54135f26292de9faf6a1043ba0817e8c2fb29cd050... | ebab12e9878344fdc93a8c1cad27ff2f709e12b90f049...  |
| 1 | 91d238c88316a444585278049cc9bdedb871414ebdb0dd... | b6954799ca47bc836ac02ab86e4710c3b0e096d7ace4...   |
| 2 | f3943bf594d541b81fbde88288600739a47ee97b0524c1... | 6ac07c3845b0ce5cac4ab3569ea70236b7cfa8fd51570...  |
| 3 | 0799215a062ddef149fa49bfe069e3ff2b4b79f08f5b01... | c67b3266ecd4af12979c1f67d1efffd0d3e9bfbcb567d2... |
| 4 | f49783f27606316e1e3a590e7403eab417b5612c2faeaf... | 1cefb8d34b92489e8ffd9cf34c469964f0527b1cae33e...  |

```
import seaborn as sns
import matplotlib.pyplot as plt
```

```
plot_df = df.melt(
    id_vars="Country of residence",
    value_vars=["sum", "PTRS"],
    var_name="Score Type",
    value_name="Score"
)

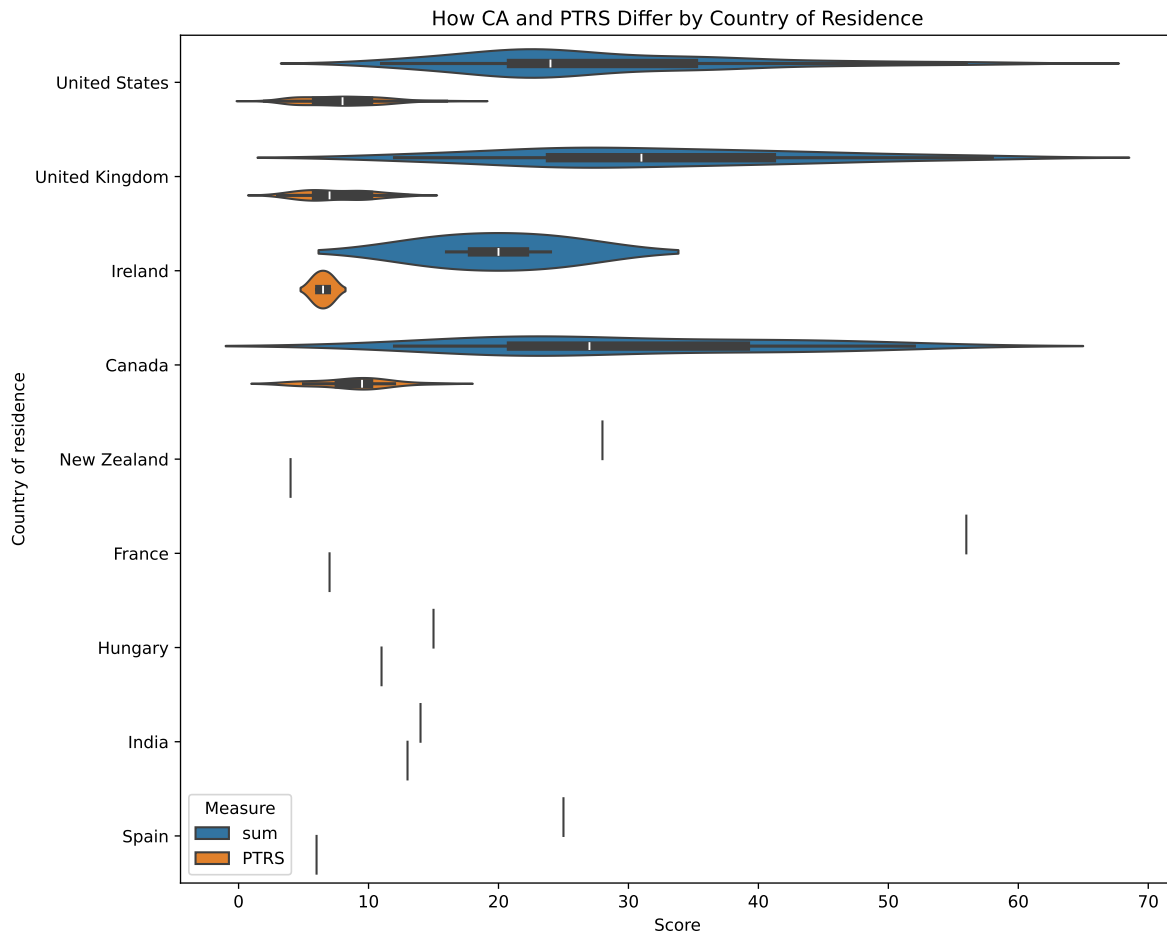
plt.figure(figsize=(10, 8))

sns.violinplot(
    data=plot_df,
    y="Country of residence",
    x="Score",
    hue="Score Type",
    split=False,
    inner="box"
)

plt.title("How CA and PTRS Differ by Country of Residence")
plt.ylabel("Country of residence")
plt.xlabel("Score")

plt.legend(title="Measure")
plt.tight_layout()

plt.savefig("figure.png", dpi=300, bbox_inches="tight")
plt.show()
```



use only the first 4 countries

```
countries_to_keep = [
    "United States",
    "United Kingdom",
    "Ireland",
    "Canada"
]

plot_df_filtered = plot_df[
    plot_df["Country of residence"].isin(countries_to_keep)
]
```

```
plt.figure(figsize=(10, 8))
```



```

sns.violinplot(
    data=plot_df_filtered,
    y="Country of residence",
    x="Score",
    hue="Score Type",
    split=False,
    inner="box"
)

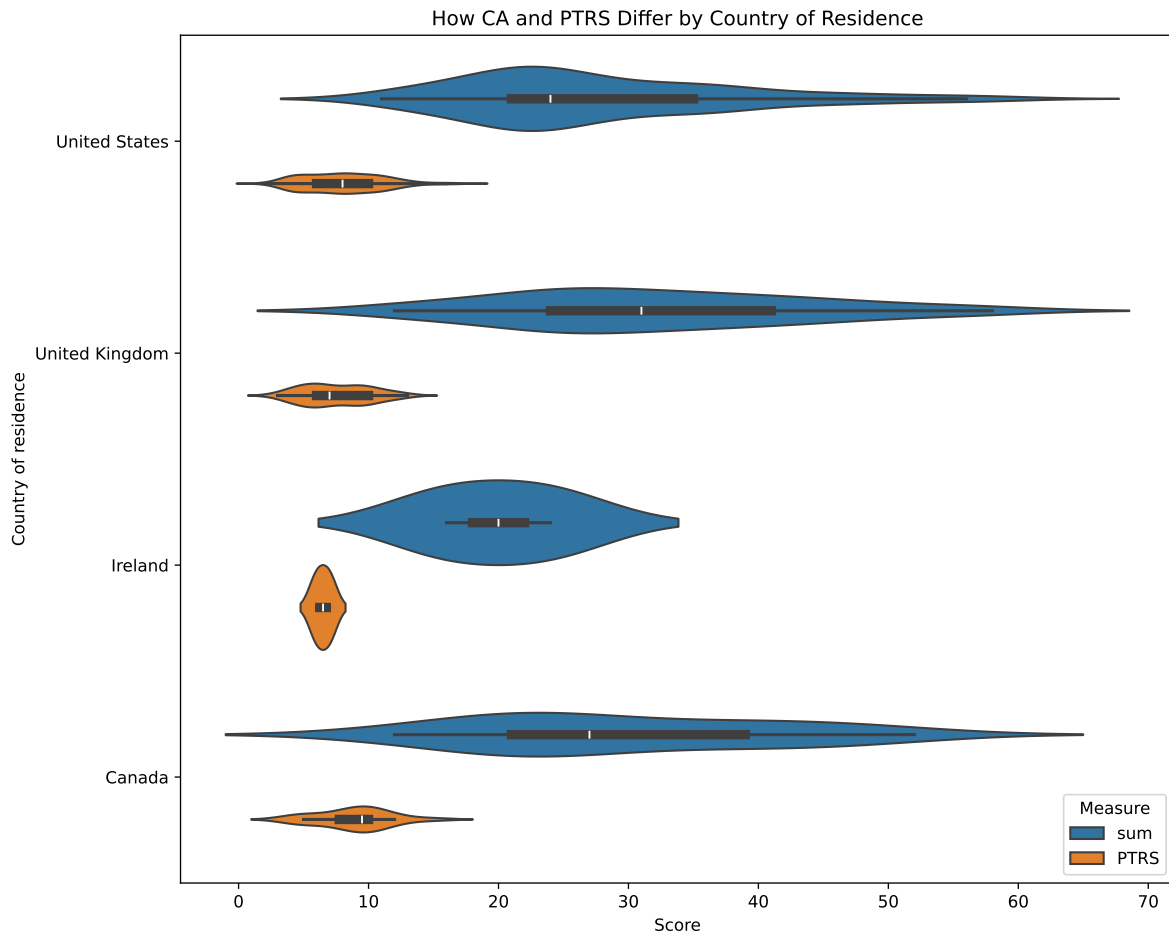
plt.title("How CA and PTRS Differ by Country of Residence")
plt.ylabel("Country of residence")
plt.xlabel("Score")

plt.legend(title="Measure")
plt.tight_layout()

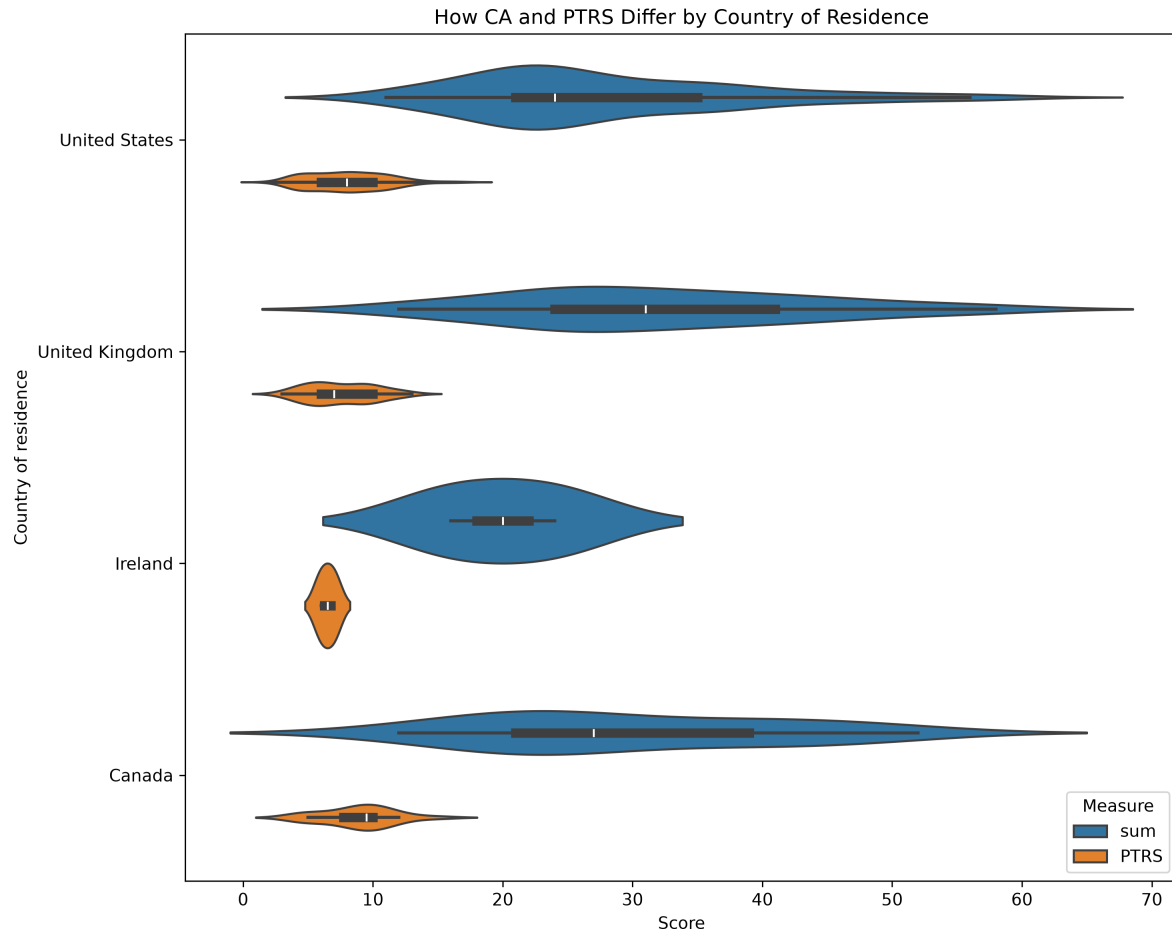
plt.savefig("figure_top4_countries.png", dpi=300, bbox_inches="tight")

plt.show()

```



## Viz



## Results

We can see from the plot in different countries, CA and PTRS have some differences.